

NDORWA SECONDARY SCHOOL
CONTINUOUS ASSESSMENT
S.6 END OF TERM 1 2026
PHYSICS 1
Time: 2 hours and 15 minutes

NAME.....

Instructions: Attempt all items in this assessment

ITEM 1

An old man was listening to 7:45 pm announcements on the voice of Kigezi, which broadcasts at 89.5MHz, while attentive, his grandson came playing a guitar(endigidi) with a string of 60cm, mass of 5g, and a tension of 100N, vibrating at its fundamental frequency. The old man asked his grandson to tell him how the two waves are different or similar. The displacement of the radio wave is given by
 $y = a \sin(100\pi t - (10/9) \pi x)$

Task: as a physics student;

- a) Help the grandson to distinguish the two waves.
- b) Using the equation of displacement above, obtain the amplitude of the resultant standing wave.
- c) Obtain the wavelengths for each of the above waves,
- d) and if the same string was made to vibrate in its 2nd overtone, what would be its velocity?
- e) Describe how stationary waves are formed to help an old man understand the phenomenon, hence state conditions for it to be formed.

ITEM 2

A team of researchers measures how much heat various substances can absorb to identify materials that maintain cooking temperatures after sunset. The team uses samples of water, sand, and specially formulated phase-change materials, heating them equally while

Do your best, and success will follow.

recording temperature changes. They discovered that water-based systems provide more consistent heating but require better insulation, while solid materials reach higher temperatures faster but cool more quickly. 2kg water and 2 kg of sand were used, each receiving 84000J of heat

$$\text{hint } [C_{\text{soil}} = 800 \text{ J kg}^{-1} \text{ }^{\circ}\text{C}]$$

Task: Using knowledge of Physics.

- a) Design an experiment that will help researchers obtain the specific heat capacity of a solid using an electrical method.
- b) If the water's temperature rises by 10 °C, calculate the temperature rise of the soil
- c) Calculate the heat energy required to raise the temperature of water from 20°C to 80°C.
- d) If equal masses of water and sand receive the same amount of heat, which will show a greater temperature increase? Explain your answer
- e) If a cylinder contained 0.5m³ of water at 200 kPa and 25 °C, obtain the number of moles of water that were used.
- f) Describe what happens to a gas when it is compressed at temperatures above and below the critical temperature.

ITEM 3

A radio repair workshop in Kampala frequently deals with faulty tuning capacitors in older radio models. The technicians need to understand how parallel plate capacitors work to diagnose and fix issues with station selection. A physics teacher at a school in Jinja uses homemade parallel plate capacitors to demonstrate energy storage principles. Students construct simple capacitors using aluminum foil sheets separated by paper dielectric materials. They measure how different dielectrics affect capacitance using multimeters and simple circuits.

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Task: as a student of physics,

- a. Derive the formula for the capacitance of a parallel plate capacitor.
- b. Explain the function of a dielectric material in a capacitor.
- c. A parallel plate capacitor has a capacitance of $10\ \mu\text{F}$ in air. Calculate its capacitance when filled with a dielectric of constant 5.
- d. Describe how the stored energy changes when a dielectric is inserted into a charged capacitor
- e. Calculate the area needed for a parallel plate capacitor with air dielectric to have a capacitance of $1\mu\text{F}$ if the plates are 1 mm apart.

ITEM 4.

A new rural health center has been built in a valley where mobile phone signals are weak. You are part of a student team challenged to design a light-based communication system between the health centre and a village watch tower located 150 meters away on a hill, in order to ease communication at the health centre. Hint: refractive indices of glass and air are 1.5 and 1.0, respectively.

Task: Using knowledge of Physics.

- a) Explain how communication occurs from the watch tower to the health center through the medium identified.
- b) Suggest **practical improvements** that the team can implement to increase clarity or strength of the signal at the health center.

END

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